

# Hybrid III 6 y/o Lab

BME254L - Spring 2026 - Palmeri

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## Tasks

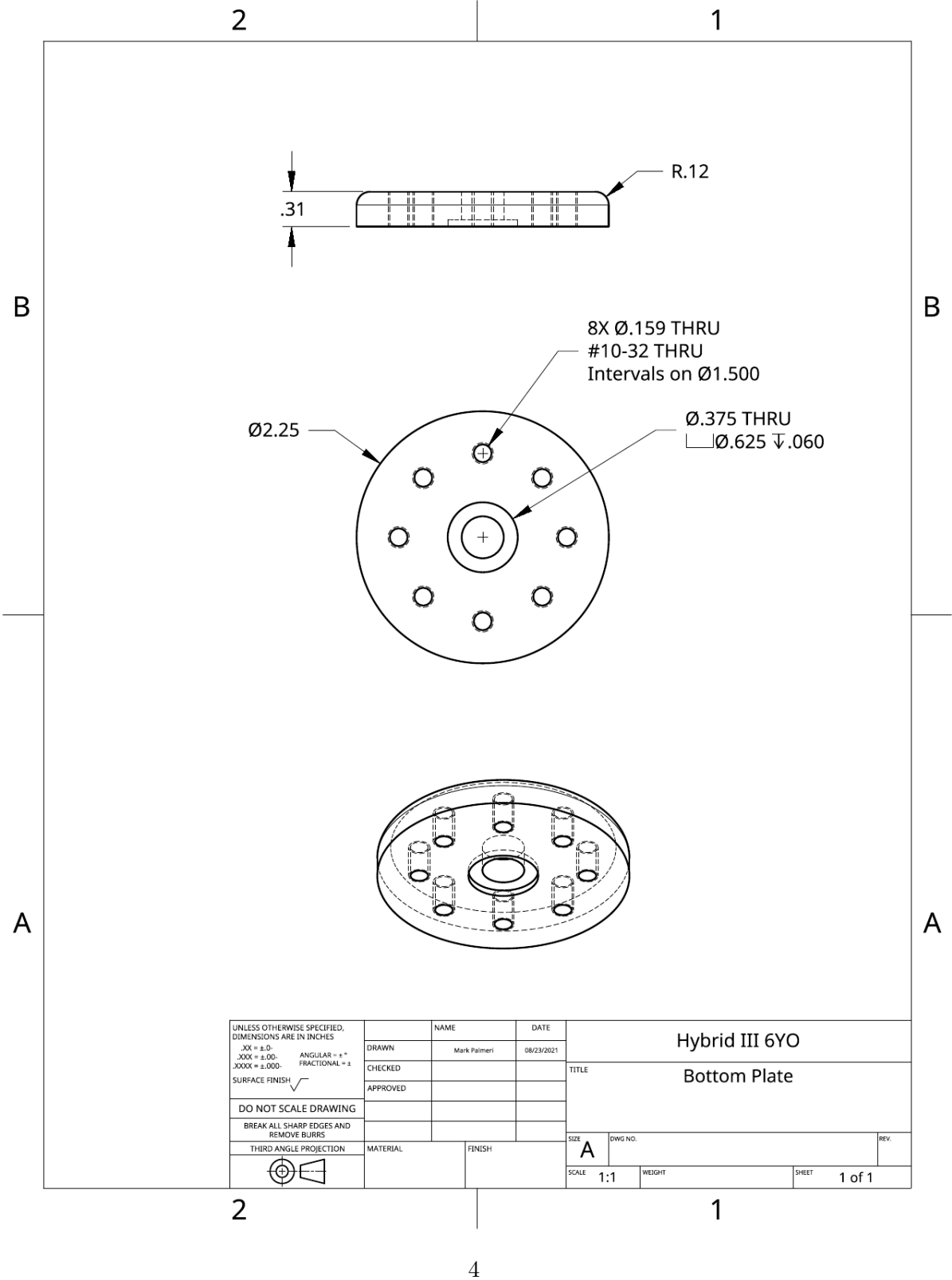
- Create all of the parts for the Hybrid III 6yo test setup based on the attached drawings.
- You should create all of these parts and their associated drawings and assemblies as part of the same “Document” and share that document with Dr. Palmeri and the TA(s).
- Create the mechanical drawings associated with each of those parts (i.e., recreate the drawings that you are using to generate the parts) and the assembly.
- Assemble your parts as guided by the assembly diagram and as shown in the uploaded animation.
- There are online tutorials on Assemblies if you need some refreshers:
  - <https://learn.onshape.com/courses/introduction-to-assembly-design>
  - <https://learn.onshape.com/courses/fundamentals-onshape-assemblies>

### **What to submit**

Upload mechanical drawings and a screenshot of the part for all of your parts to the Hybrid III 6 y/o Canvas assignment.

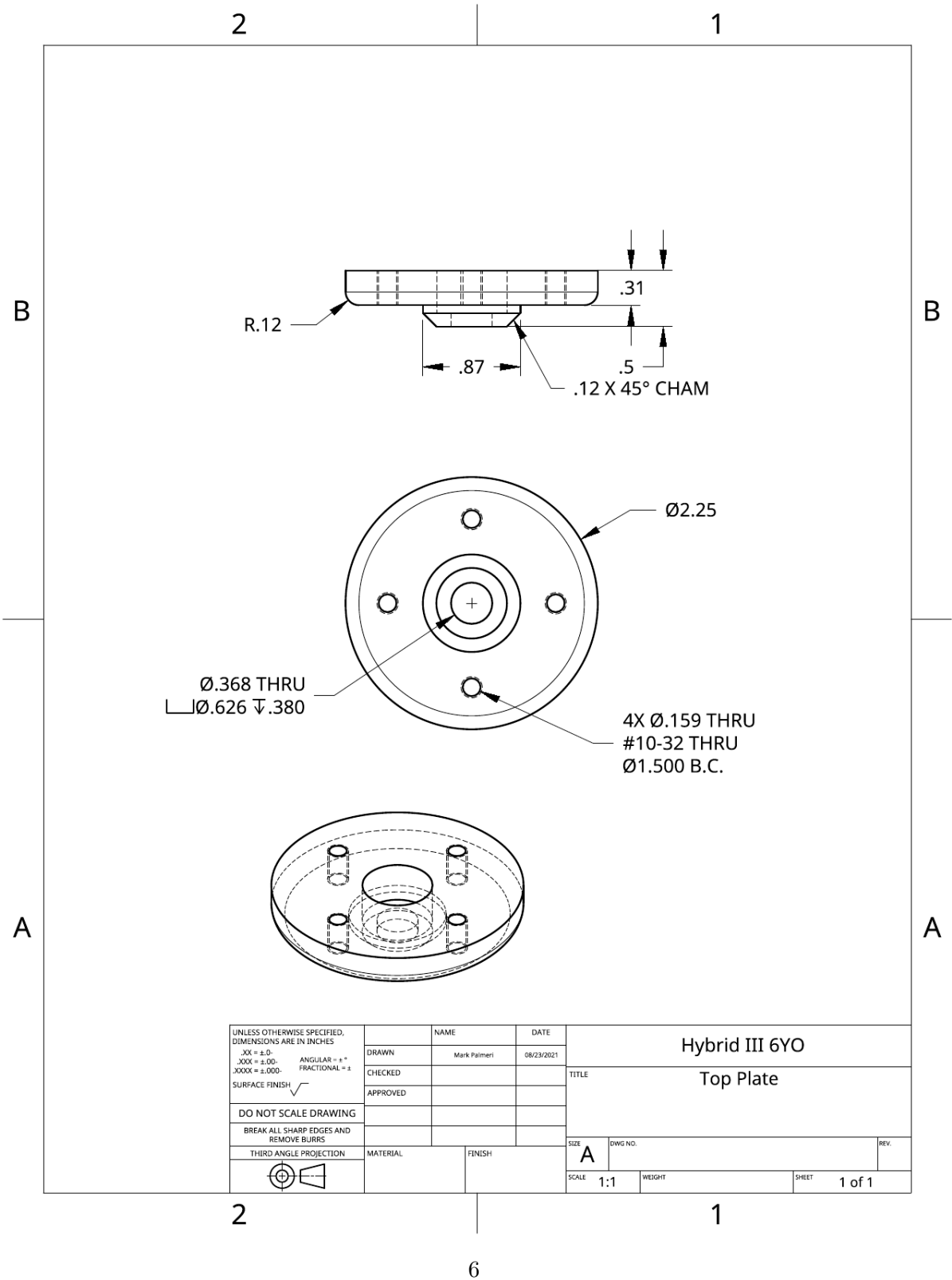


Bottom Plate



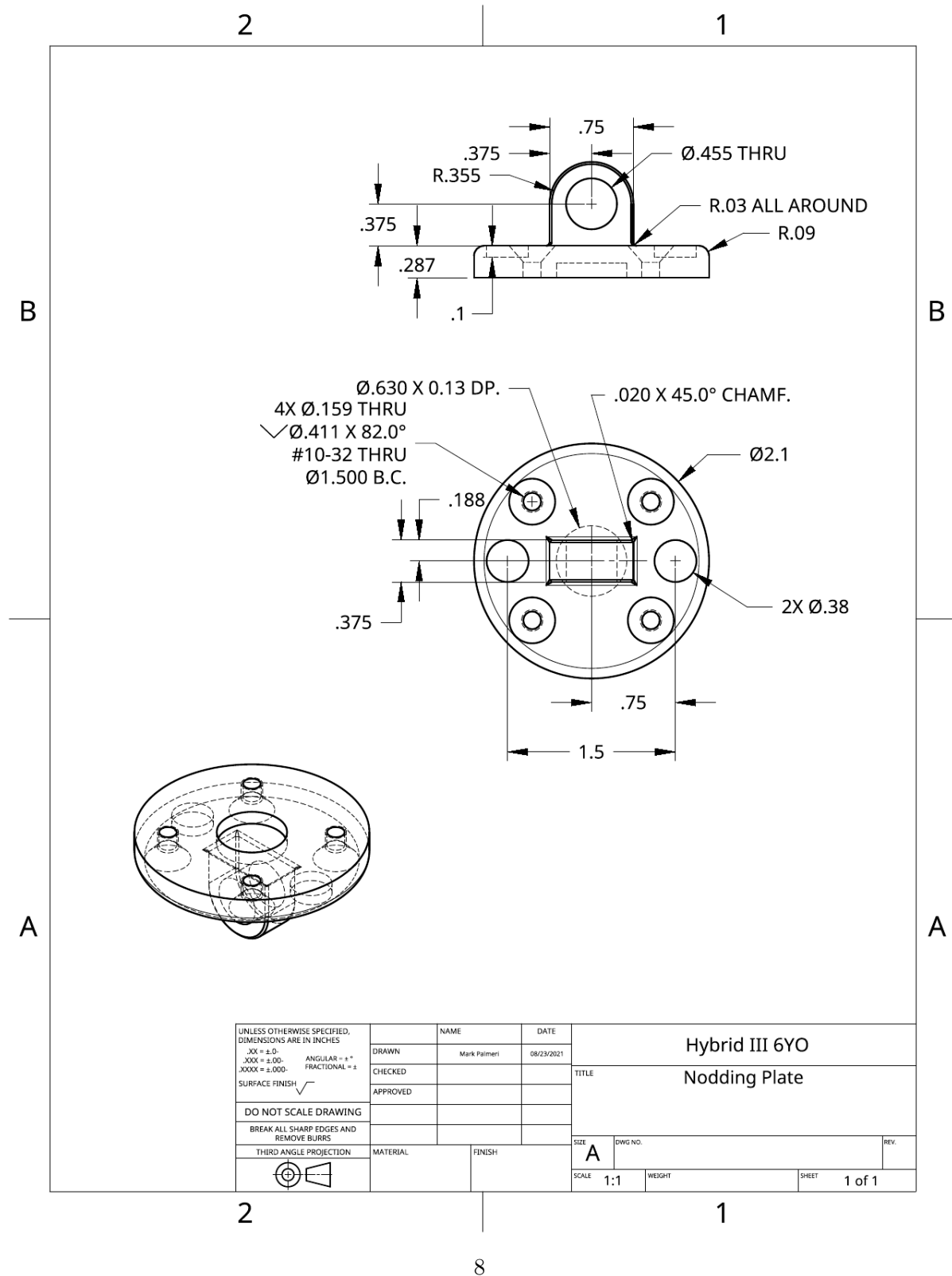


Top Plate





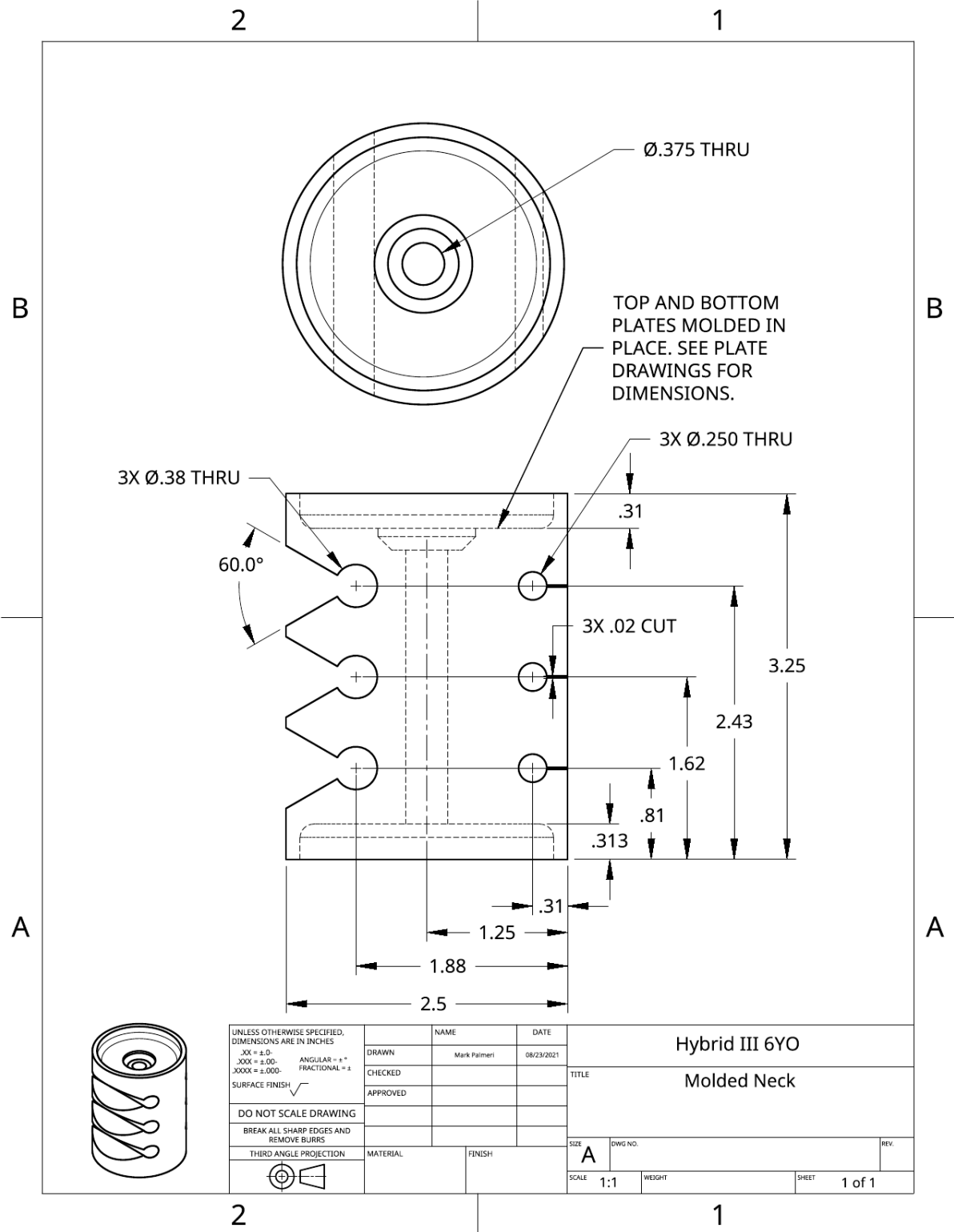
## Nodding Plate







Molded Neck

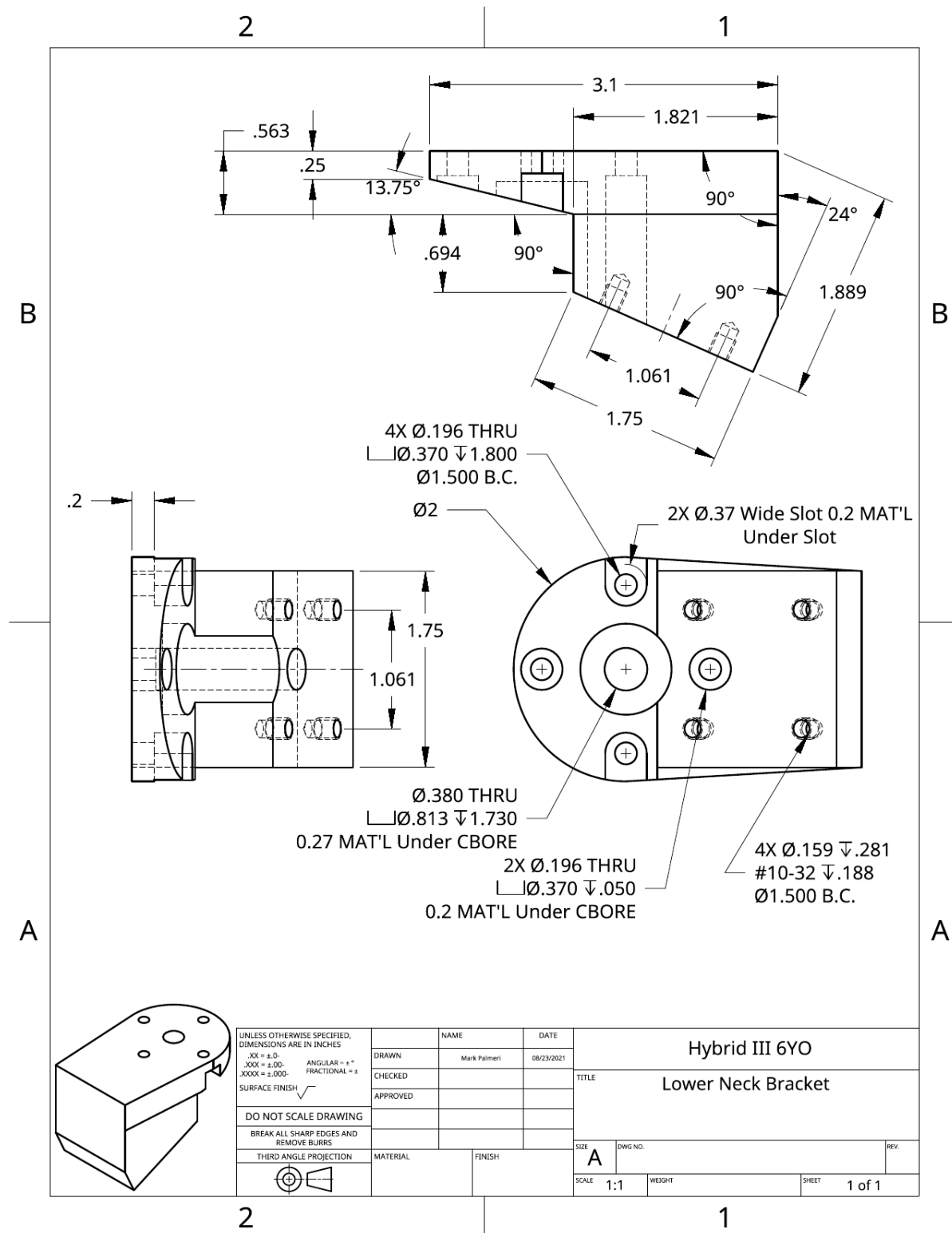


 Tip

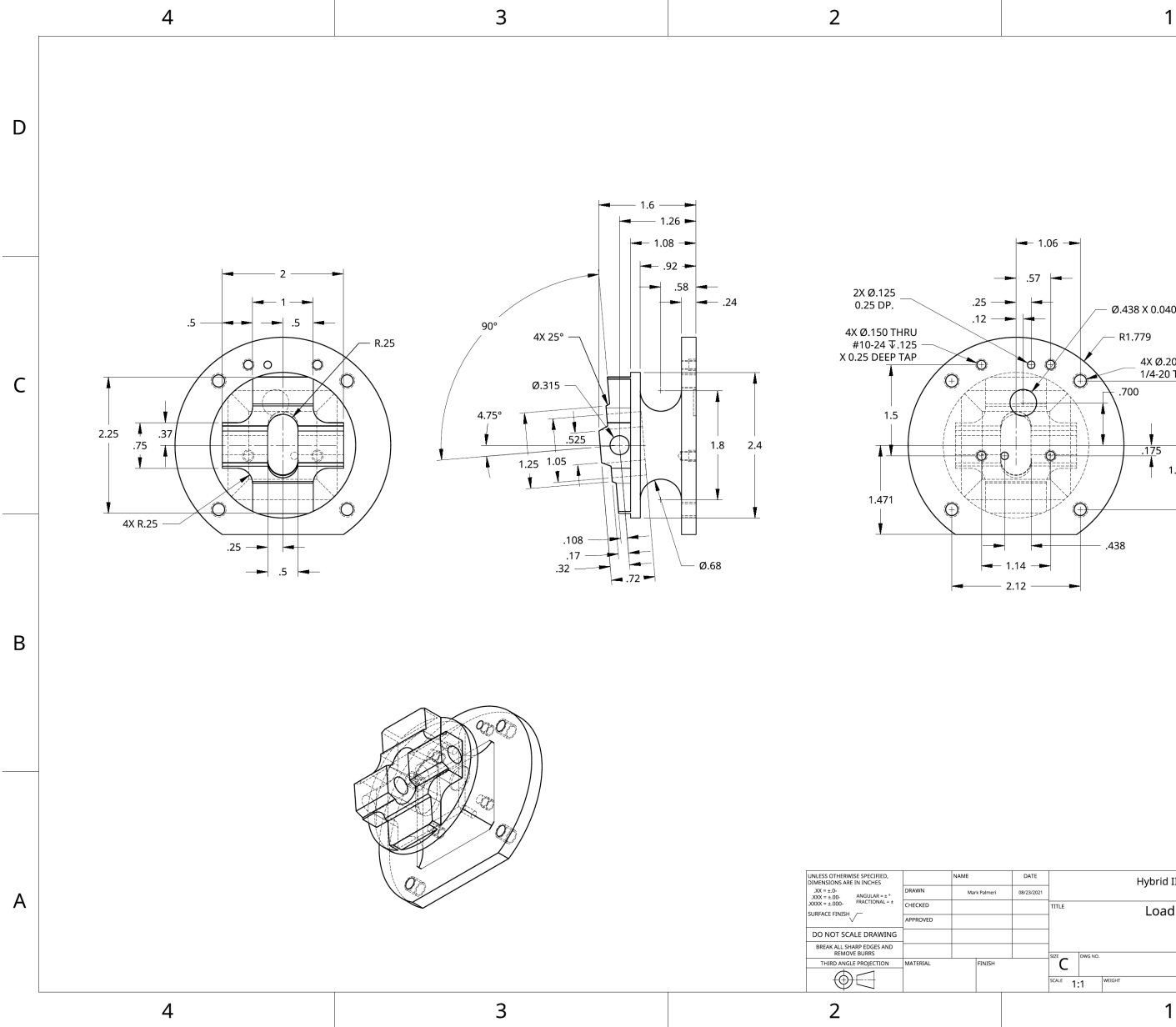
Note that the this part has features that are dependent on the top and bottom plates. To reduce the overhead of manually entering these dimensions from the other parts, you can use the **Derived** tool in Onshape to bring in the relevant faces from the other parts. <https://cad.onshape.com/help/Content/derived.htm#StepsDesktop>



# Lower Neck Bracket



Load Cell



SECTION A - A  
SCALE 1:2

SCALE 1:2

SCALE 1:2

| Item | Quantity | Part number        | Material |
|------|----------|--------------------|----------|
| 1    | 1        | MOLDED NECK        | Butyl    |
| 2    | 1        | TOP PLATE          | Steel    |
| 3    | 1        | NODDING PLATE      | Steel    |
| 4    | 1        | BOTTOM PLATE       | Steel    |
| 5    | 1        | LOWER NECK BRACKET | Aluminum |

UNLESS OTHERWISE SPECIFIED,  
DIMENSIONS ARE IN INCHES:  
 .XX = ±.01  
 .XXX = ±.005  
 .XXXX = ±.0005  
 ANGULAR = ±.5°  
 FRACTIONAL = ±.005  
 SURFACE FINISH: 

DO NOT SCALE DRAWING  
 BREAK ALL SHARP EDGES AND  
 REMOVE BURRS

THIRD ANGLE PROJECTION

MATERIAL: \_\_\_\_\_ FINISH: \_\_\_\_\_

NAME: \_\_\_\_\_ DATE: \_\_\_\_\_

DRAWN: Mark Palmeri 08/23/2021

CHECKED: \_\_\_\_\_

APPROVED: \_\_\_\_\_

TITLE: Hybrid III 6YC Assembly

SIZE: B DWG NO. \_\_\_\_\_

SCALE: 1:1 WEIGHT: \_\_\_\_\_